

Environmental Evidence

What is the evidence for the impact of ocean warming on subtropical and temperate corals and coral reefs? A systematic map.

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Manuscript Number:	ENEV-D-24-00044
Full Title:	What is the evidence for the impact of ocean warming on subtropical and temperate corals and coral reefs? A systematic map.
Article Type:	Systematic Map
Abstract:	Abstract Background. Subtropical coral reefs are comparatively understudied compared to tropical reefs, yet host a diverse and abundant array of marine life and provide substantial socio-economic benefits to communities. Research into the impacts of climate change on subtropical coral reefs increased over the past two decades as bleaching events and increasingly frequent degradation events by ocean warming affect these systems. Understanding the extent of research effort and type of evidence assessing the response of subtropical corals and reefs to climate change can provide valuable insight into global patterns in research effort, and identify critical knowledge gaps that will impact our ability to predict and respond to future changes in subtropical coral reefs. Here a systematic map approach is used to identify research efforts from 2010 to 2023 and highlight patterns in the type, scale, and location of research conducted and the availability of data reported. Methods. Primary literature within Scopus and Web of Science databases were identified and extracted. The methodologies outlined in a previously published systematic map protocol were applied, and 91 primary research publications were identified for data extraction. Data extraction included bibliometric data, discipline and type of research, type of data reported and how it was recorded, and data availability. Findings. The identified literature consisted primarily of experimental (46%) and observational (38%) studies. The majority of the primary literature investigated corals in the ecoregion of Southern China (13%), Western Mediterranean (10%) and a total of seven ecoregions within Oceania (29%). Stressors reported as drivers of ocean warming reflect standardisation of event reporting in literature. Standardised metrics related to degree heating week (DHW) and marine heatwave (MHW) schemes are increasingly reported, particularly in Australia. Finally, the need for research in regions such as the Western Indian Ocean is also clear, as the number of studies highlighting evidence of ocean warming on the system they provide are less than their counterparts in other subtropical ecoregions. Conclusions. Climatic changes, particularly those associated with increasing ocean temperatures, and its effects on subtropical and temperate coral reefs are of increasing concern to policy makers and researchers. This systematic map provides a broad overview of research topics and effort around the globe since 2010, and identifies areas where more research efforts are needed. It identified the major research clusters in Asia, Australia, Mediterranean, and North America, gaps of research in regions such as East Indian Oceans. Without attention to these gaps, local management decisions may be based on information from other regions, and evidence from other regions may not apply to these local regions.

What is the evidence for the impact of ocean warming on subtropical and temperate corals and coral reefs? A systematic map.

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Abstract

Background.

Subtropical coral reefs are comparatively understudied compared to tropical reefs, yet host a diverse and abundant array of marine life and provide substantial socio-economic benefits to communities. Research into the impacts of climate change on subtropical coral reefs increased over the past two decades as bleaching events and increasingly frequent degradation events by ocean warming affect these systems. Understanding the extent of research effort and type of evidence assessing the response of subtropical corals and reefs to climate change can provide valuable insight into global patterns in research effort, and identify critical knowledge gaps that will impact our ability to predict and respond to future changes in subtropical coral reefs. Here a systematic map approach is used to identify research efforts from 2010 to 2023 and highlight patterns in the type, scale, and location of research conducted and the availability of data reported.

Methods.

Primary literature within Scopus and Web of Science databases were identified and extracted. The methodologies outlined in a previously published systematic map protocol were applied, and 91 primary research publications were identified for data extraction. Data extraction included bibliometric data, discipline and type of research, type of data reported and how it was recorded, and data availability.

Findings.

The identified literature consisted primarily of experimental (46%) and observational (38%) studies. The majority of the primary literature investigated corals in the ecoregion of Southern China (13%), Western Mediterranean (10%) and a total of seven ecoregions within Oceania (29%). Stressors reported as drivers of ocean warming reflect standardisation of event reporting in literature. Standardised metrics related to degree heating week (DHW) and marine heatwave (MHW) schemes are increasingly reported, particularly in Australia. Finally, the need

for research in regions such as the Western Indian Ocean is also clear, as the number of studies highlighting evidence of ocean warming on the system they provide are less than their counterparts in other subtropical ecoregions.

Conclusions.

Climatic changes, particularly those associated with increasing ocean temperatures, and its effects on subtropical and temperate coral reefs are of increasing concern to policy makers and researchers. This systematic map provides a broad overview of research topics and effort around the globe since 2010, and identifies areas where more research efforts are needed. It identified the major research clusters in Asia, Australia, Mediterranean, and North America, gaps of research in regions such as East Indian Oceans. Without attention to these gaps, local management decisions may be based on information from other regions, and evidence from other regions may not apply to these local regions.

Keywords: Coral; Coral Reefs; Scleractinian; High Latitude; Marginal; Climate Change; DHW; MHW

Background

Coral reef ecosystems are amongst the most important ecosystems in the world, where unique marine biodiversity provides a range of socio-economic benefits to communities [1]. However, the condition of coral reefs have been assessed as declining globally under the effects of climate change [2, 3] threatening the biodiversity of marine environments, and even leading to local extinction of species [4]. Coral species are found in a range of locations, from solitary corals in temperate regions to the dramatic reefs of the tropics. Tropical coral reefs have received most attention and long-term studies have shown climate-related declines in number of species, highlighting the consequences of degradation of coral reefs. These issues are therefore pressing and researchers around the world have made significant efforts to identify the issues, propose solutions, and increase public awareness [6]. Despite these efforts, tropical coral reef conditions around the world continue to decline [7, 8]. The continued increase in ocean temperature causes bleaching and mortality in tropical reefs and the expansion of tropical coral species into subtropical regions [9, 10]. In 2010, bleaching events on subtropical coral reefs in Hong Kong [11, 12] and Japan [13] indicated that ocean warming is also affecting subtropical marine ecosystems. These bleaching events in recent decades reflect the need to assess the state of subtropical coral reefs to conserve them. The subtropical coral reefs differ from tropical coral reefs in terms of variety of species and environment [14], whilst also serving as habitat to a variety of subtropical and tropical marine organisms, including a number of endemic taxa [15]. Moreover, studies have also reviewed the capability of these coral reefs in subtropical regions to become refugia for tropical species [16]. This pattern has led to new research, and more efforts directed to the conservation of subtropical coral reefs.

Despite the research efforts addressing the conservation of subtropical coral reefs, there are contradictions between studies investigating the impacts and future for these ecosystems, from acting as a refuge from climate change [17, 18], to accommodating poleward expansion

of tropical species [18, 19], to suffering decline and loss of regional endemism [20]. The conditions for a site to act as a coral refugia depend largely on the surrounding oceanographic conditions, for example, upwelling can protect thermally sensitive coral genera on the forereef of the subtropical atoll by decreasing the temperature of water [16, 17]. However, the cool upwelled water will only provide refuge if there is synchrony between timing of ocean warming and upwelling [17]. Equally, studies have shown that under climate change herbivorous marine organisms and corals from tropical regions are expanding into subtropical and temperate regions, and these expansions have the potential to alter the community structure of subtropical and temperate regions [18, 21]. This influx of tropical poleward expansion could pose a threat to the endemic subtropical and temperate coral communities, as some tropical coral species such as *Acropora hyacinthus* and *Acropora muricata* have much higher growth rates than other endemic species in these regions [22]. The endemic taxa may therefore face decline as the result of competition tropical species [18]. With the increase in number of studies on the topic, there is a need to compile existing evidence into a systematic map to provide a summary of the current state of knowledge and identify knowledge gaps.

Here we describe the use of a systematic map to provide an overview of distribution and abundance of evidence in relation to a broad research topic [23]. Such evidence maps enable the identification of gaps in research, knowledge clusters, study themes, and methodologies in studies [24]. Additionally, analyses of bibliometric data such as citation numbers, publishing information and author collaboration details from published studies reveal patterns which can be used to identify the research specialties, research groups and impact of studies. The insights provided by a systematic map can therefore help direct future research efforts more effectively.

When collating and classifying research evidence, factors such as the location of the study, year of publication, study types, and metric used for identifying ocean warming events can be classified. With regard to location, the Marine Ecoregions of the World (MEOW) [25] introduced a hierarchical biogeoregionalisation of coastal and shelf areas based on

130 biogeographic assessments, ecoregional assessments, government-derived or supported
131 systems and input from different researchers and assessments [25]. Using MEOW as a
132 standardised reference for location enables a systematic approach for recording locational
133 data in marine conservation planning and research, fostering a more robust referencing
134 between studies to identify biogeographical data. In the early 2010s, more publications began
135 to record the effects of ocean warming and extreme weather events on subtropical coral reefs
136 [26-28]; therefore, assessing studies from 2010 onwards will reveal the knowledge gained on
137 this topic. While many methods of recording extreme temperature events are used; most
138 common methods of recording include degree heating weeks (DHW) [29-31] and marine
139 heatwaves (MHW) [32]. Using standardised metrics to assess heating events allows direct
140 comparison between studies. Identifying the literature that utilised these standardised
141 methods will provide insights into the adoption of using DHW and MHW in study of subtropical
142 coral reefs.

Objective of the Review

The following questions outlined in the published protocol [33] will be answered in this systematic map:

What is the evidence, within scientific literature since 2010, on the impact of ocean warming to subtropical and temperate corals and coral reefs?

How have sea surface temperature (SST) data been recorded in subtropical and temperate coral reef locations?

Are there studies using MHW (Marine heatwaves) [32] and/or DHW (degree heating weeks) [29-31] as preferred metrics when evaluating climate events?

How are the levels of mortality and bleaching defined in the research locations?

Where, and how, have coral biological response and coral reef ecological impacts been recorded in subtropical and temperate coral reef locations?

The questions listed above concern the rording methods of existing literature as well as the distribution of researchers around the world working on the topic. Through answering these questions, trends in the use of different temperature data, reporting metrics and publication timeline can be identified, as well as providing a summary of recent publications.

Terminology.

Ocean warming. Ocean warming refers to increase of temperature in oceanic waters.

Coral. Members of the Anthozoa orders; Scleractinia, Corallimorpharia, Alcyonacea and Antipatharia which include the true stony corals, soft corals, black corals, etc.

Coral bleaching. Coral bleaching occurs when the water that the coral system lives in becomes warmer than the upper temperature sensitivity threshold of the coral, causing coral to expel the symbiont zooxanthellae, leaving a white coral skeleton.

Coral populations. Mixed and single species populations of coral within an ecosystem.

Coral reef ecosystem. Coral dominated ecosystem.

Coral reef refugia. Coral reef refugia are areas where the physical, biological, and ecological characteristics allow for a potential habitat for corals sheltered from the suboptimal conditions resulting from climate change [34].

Degree Heating Weeks. Degree heating weeks (DHWs) is a measure of accumulated thermal stress experienced by corals. It is calculated by adding up days where temperatures exceed the usual summertime maximum by at least 1 degree Celsius over a 12-week period [35].

Ecoregions. Defined by Spalding et al. (2007) [25] as areas of relatively homogeneous species composition and clearly distinct from adjacent systems. Ecoregions are nested within Provinces.

High latitude reef. Reefs that are in regions above and below 28°N and 28°S respectively [36]. As referred in Beger et al. [37] work and the henceforth figure and table (Figure 3 And Table. S4, see appendix).

Marine Heatwave. Marine heatwave (MHW) descriptions are calculated from long-term sea surface temperature (SST) data obtained by satellite, in-situ instruments, or other means of

observed data collection. A MHW is defined as a thermal event where SST exceeds the 90th percentile of the local climatological temperature for at least 5 consecutive days [32].

Marine Protected Areas. Marine Protected Areas (MPAs) are defined as areas with a clearly defined geographical space, recognised, dedicated, and managed through legal or other means to maintain the long-term conservation of ecosystems and cultural values associated with the location [38].

Marginal reef. Marginal reefs are where the environmental conditions are marginal or close to a threshold for the survival of the coral species, meaning that the living conditions are less than ideal but still feasible for coral species to survive [39].

PECO. (P)opulation, (E)xposure (C)omparator and (O)bjective. It is a framework used to define a problem statement in exposure studies, originally derived from a framework used in medical studies of interventions (PICO).

Primary literature. These are literatures where original results from the authors were reported in the form of a report of their findings.

Provinces. Defined by Spalding et al. (2007) [25] as large areas with presence of distinct biotas, that hold some level of endemism.

Sea Surface Temperature. Sea surface temperature (SST) is the measure of the surface temperature of ocean.

Secondary literature. These are literatures in which authors synthesize information from primary literature into a separate document, such as a review.

Subtropical. The region surrounding the tropical regions. However, in this work the subtropical ecoregions are defined as in Figure 3 and Table S1 (See appendix) [25, 37] and interchangeable with high latitude reefs.

Systematic map. Methodical overview of the quantity and quality of evidence in relation to an question of policy or management relevance [40].

Temperate. The region surrounding the subtropical regions. However, in this work the temperate ecoregions are defined as in Figure 3 [37].

Tropicalisation. Tropicalisation refers to a region outside the tropical latitudes that is increasing in the number of warm-affinity species and decreasing in the number of cool-water species [21].

212 **Methods**

213 **Protocol**

214 We followed the Collaboration for Environmental Evidence Guidelines and Standards for
215 Evidence Synthesis in Environmental Management [40] and conformed to the reporting
216 standards for systematic evidence synthesis (ROSES) [41]. The systematic map is based on
217 a previously published protocol registered in PROCEED, the global database of prospectively
218 registered evidence reviews and syntheses in the environmental sector [33].

219 **Deviation from the protocol**

220 There was no deviation from the protocol.

221 **Search strategy**

222 10 benchmark articles were selected a priori for the development of the search string. These
223 10 articles were selected based on their relevance to the topic, covering both climatology and
224 subtropical and temperate coral systems under the influence of ocean warming. A series of
225 search strings were trialed to test their capability to capture the benchmark articles. The
226 Boolean operator search string was first developed on Scopus and translated to the Web of
227 Science database after it successfully captured all 10 benchmark articles on Scopus. To
228 capture recent studies, the search results were limited to publication on and after 2010. The
229 search was  limited to publications in English only.

Search strings

Bibliographic databases

Scopus: TITLE-ABS-KEY ((coral* OR "coral reef*")) AND TITLE-ABS-KEY (("ocean warming" OR "marine heatwave*" OR "marine heat wave*" OR mhw* OR "degree heating week*" OR dhw* OR "heat stress*" OR stress OR tropicali?ation OR temperature* OR "climate change")) AND TITLEABS- KEY ((mortal* OR surviv* OR health* OR diseas* OR grow* OR reprodu* OR cover* OR tropicali?* OR shift* OR habitat* OR increase* OR declin* OR decreas* OR impact* OR threat* OR bleach* OR acclimati?* OR respons*)) AND TITLE-ABS-KEY ((marginal OR "high latitude" OR temperate OR subtropic* OR extratropic*)) Followed by a removal of articles published prior to 2010.

Web of Science: (All Fields) ((coral* OR "coral reef*")) AND (All Fields) (("ocean warming" OR "marine heatwave*" OR "marine heat wave*" OR mhw* OR "degree heating week*" OR dhw* OR "heat stress*" OR stress OR tropicali?ation OR temperature* OR "climate change")) AND (All Fields) ((mortal* OR surviv* OR health* OR diseas* OR grow* OR reprodu* OR cover* OR tropicali?* OR shift* OR habitat* OR increase* OR declin* OR decreas* OR impact* OR threat* OR bleach* OR acclimati?* OR respons*)) AND (All Fields) ((marginal OR "high latitude" OR temperate OR subtropic* OR extratropic*)) Followed by removal of articles published prior to 2010.

Web-based search engines

Grey literature was searched on Open Access Theses and Dissertations (OATD) database, with the following search string used, which in pilot testing has yielded 8 results: coral AND (subtropical OR temperate) AND "ocean warming". The included results were limited to articles and reports published since 2010 to cohere with the rest of the searches. Grey literature included was limited to research theses, pre-review reports, and open access scientific studies (herein defined as grey literature selected for inclusion in systematic

mapping) while government reports were excluded due to the inconsistency in availability of these reports between regions.

Comprehensiveness of the search

All 10 manually collected benchmark articles were returned in the Scopus search, ensuring the search string to be sensitive enough for this systematic mapping study. The list of the 10 benchmark articles is found in Additional file 1.

Screening strategy

The first screening was conducted including and excluding articles by titles and abstracts using Rayyan software [42] and was undertaken by two reviewers independently on 1,218 titles and abstracts. Articles included in the first round of screening were screened again in the second phase of the screening by full text. Where full text did not meet the selection criteria, they were excluded from the data extraction. A pilot screening was performed by two of the authors to assess the coherence of articles screening agreement by Cohen's Kappa coefficients [43, 44]. Articles on Rayyan were selected at random for reviewers to quickly screen through the titles and abstracts (30, 218, 24.6%). Articles with uncertainty in decisions were flagged for a third independent reviewer (TDA) for final decision. Consistency checks were performed for both stages on 30 titles and abstracts (Kappa = 0.61). Full text data inclusion and exclusion were cross checked by the author (MLH) to ensure that all data extracted are coherent to the inclusion criteria.

Eligibility criteria

Articles from the databases were screened manually according to the following PECO criteria:

Population: Subtropical and Temperate Coral Systems and Coral Reefs

- Studies focused on or including any subtropical and temperate coral systems and coral reefs were included; this includes marginal and high latitude coral systems.

- Studies focused solely on tropical, polar and/or deep sea corals, kelp forests and fishes were excluded.

Exposure: Ocean Warming

- We included studies focused on the effects of exposure to ocean warming, specifically empirical ocean warming studies that utilise degree heating weeks and the definition of marine heatwaves as defined by Hobday et al. (2016) [32], where:
 - SST exceeds the bleaching threshold of 4 degrees Celsius.
 - SST exceeds the 90th percentile of the local climatological value for the time of year based on 30 years of climatology data [32].
- Studies focused on ocean acidification and its interaction with ocean warming were excluded from this systematic map.

Comparator: Effects before and after recorded event/exposure.

Outcomes: Mortality and bleaching, habitat changes, threats, acclimatisation of subtropical and temperate coral systems.

- We included all studies that address the physiological effects and ecological effects ocean warming have on the coral systems and reefs – this includes mortality, bleaching, changes in habitats (e.g., destruction of reef structures), threats (e.g., increase in species that pose threats to corals and coral reefs). Topics addressing the acclimatisation of species were also considered.

Reporting screening outcomes

During the full-text screening, studies that were rejected from the process and the respective reasons were recorded. A flow diagram has been used to record and visualise the number of studies assessed and rejected at each stage.

Study validity assessment.

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3 303 No study validity assessment of studies has been performed for this systematic map.

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6 304 **Data coding strategy**

7 305 Data extraction was done collaboratively by 6 reviewers using a questionnaire on the articles
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9 306 included in the full text screening. The following categories of variables were extracted and
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11 307 coded in a spreadsheet (Additional file 2):

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14 308 - Bibliographic map (Title, DOI, number of citations, authors' information, year of
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16 309 publication, journal, keywords)
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18 310 - Location of research (Research ecoregions, type of location, development status,
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20 311 species)
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22 312 - Research data recorded (Temperature data recording methods, study type, study
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24 313 timeline, environmental parameters, availability of temperature data)
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26 314 - Ocean warming evidence (Extreme weather events recorded, stressors, severity,
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28 315 extreme weather events timeline, study outcome or prediction)
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33 316 **Demonstrating procedural independence**

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36 317 As some of the authors of this proposed review may have been authors of some of the studies
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38 318 to be reviewed in this work, their studies were independently assessed by other reviewers
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41 319 without discussion with the study authors to ensure the independence of the review.
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Reporting. Methods outlined in previously published protocol were followed, apart from minor additions and deviations described in the section  "Deviation, additions and justifications".

The ROSES Reporting Standard for the reports of findings from the systematic map was applied [41].

All figures on this systematic map were created using R version 4.2.2 [45] on R Studio version 2023.06.0 [46], as well as RAWGraphs [47]. Several packages in R were used in the creation of figures, including ggplot2 (3.4.4) [48], wordcloud (2.6) [49], wordcloud2 (0.2.1) [50], tidyverse (2.0.0) [51], dplyr (1.1.4) [52], reshape2 (1.4.4) [53], viridis (0.6.5) [54], RColorBrewer (1.1.3) [55], leaflet (2.2.1) [56], htmlwidgets (1.6.4) [57], webshot (0.5.5) [58], and tm (0.7.11) [59]. Organisation of plots into multi-panel figures were done with the use of Adobe Illustrator 2024 (Version 28.5, Adobe).

Review findings

Data characteristics. From the 1,236 initially identified studies, 91 studies met the inclusion criteria for the purpose of this systematic map (Figure 1). All 91 studies provided data for the formulation of the systematic map and bibliometric map.

Study types. 44 studies were identified (48.4%) that were observation-based, three studies (3.3%) included both observational and modelling evidence. 35 studies (38.5%) were identified as experimental, with 7 studies (7.7%) included both experimental and observational evidence and one study (1.1%) included both experimental and modelling evidence. One study (1.1%) identified as solely modelling based. The top three article keywords used in included studies were “coral”, “change coral” and “bleaching coral” respectively (see a word cloud of all key words in Figure 2a).

Bibliometric map. Increasing research effort from 2013 to 2015 is reported (a total of 19 publications during 2013 - 2015) following by a drop to four publications in 2016 and increase to nine publications in 2017 (Figure 2b), and has reached a peak of 10 publications per year in 2022 and Mid-2023. In terms of number of citations, two studies stood out, having 510 citations (study by Yamano et al., 2011 [18]) and 159 citations (study by Le Nohaïc et al., 2017 [60]), respectively. Other publications included in this systematic map are less cited (Median = 20) (Figure 4a). Last authors with more than one publication are associated with institutions in Hong Kong (three authors), Australia (one author), United States of America (one author), United Kingdom (one author), Italy (one author), Taiwan (one author), Spain (one author), New Zealand (one author), and Monaco (one author). In this systematic map, we took the assumption that the last author is a senior author, however, we acknowledge that it may not be the case for all fields for literature included in this map. Most last authors are publishing only intermittently (Figure 5). Figure 6 shows the respective last authors and the location of research included publications focused on. The leading locations are in Southern China, with P. Ang having the highest number of publications focusing on the area. This is followed by numerous research in Western Mediterranean, with authors J. Garrabou and S. Goffredo, both

358 sharing the same number of publications on the area. Lord Howe and Norfolk Islands came
359 third in terms of number of last authors and their focus on the area, with S. Davy having the
360 highest number of publications in the area.

361 In terms of all the authors regardless of position of first, last, or other co-authors, P. Ang has
362 the highest number of publications (Seven publications) in the literature included for this
363 systematic map, followed by C. Ross with five publications. A list of authors, including A. Chui,
364 X. Huang, C. Linares, M. McCulloch, J.W. Qiu, R. Tsang, and K. Yu, with four publications
365 each (all extracted author names with the counts of their included works are shown in Table
366 1).

367 In terms of publication journals, the largest number of studies included come from Coral Reefs
368 (17 studies), followed by Frontiers in Marine Science (seven studies), Scientific Reports (six
369 studies) (Figure 4b). Other journals include Science of the Total Environment (four studies),
370 Diversity (three studies), Global Change Biology (three studies), Marine Ecology Progress
371 Series (three studies), Marine Pollution Bulletin (three studies), PloS One (three studies),
372 Proceedings of the Royal Society B: Biological Sciences (three studies), Bulletin of Marine
373 Science (two studies), Ecology and Evolution (two studies), Journal of Experimental Marine
374 Biology and Ecology (two studies), Limnology and Oceanography (two studies), Molecular
375 Ecology (two studies) (Table 2 with all included articles).

Location of research.

The provinces were identified, and subsequently the ecoregion where the included studies were conducted (Figure 7). There were 6 studies that involve both tropical and subtropical ecoregions in Australia. Majority of studies were conducted in the province of South China Sea (20 occurrences), this includes the ecoregions of Southern China (15 studies), South China Sea Oceanic Islands (one study), and Gulf of Tonkin (four studies). The second highest province is Mediterranean Sea, consisting of 15 occurrences, including the ecoregions of Western Mediterranean (11 studies), Adriatic Sea (two studies), Levantine Sea (one study), and Ionian Sea (one study). The province of Warm Temperate Northwest Pacific was the third highest occurring province, with 12 occurrences of studies identified in the ecoregions of East China Sea (six studies), Central Kuroshio Current (six studies) (Figure 7b).

When examining the provinces recorded from the included studies, seven provinces that are on the Australian continent were identified, including Lord Howe and Norfolk Islands (nine occurrences), Southwest Australian Shelf (six occurrences), West Central Australian Shelf (five occurrences), East Central Australian Shelf (four occurrences), Northeast Australian Shelf (three occurrences), Southeast Australian Shelf (two occurrences) and Northwest Australian Shelf (one occurrence). These provinces include the ecoregions of Lord Howe and Norfolk Island (nine studies), Leeuwin (six studies), Houtman (five studies), Tweed-Moreton (three studies), Manning-Hawkesbury (one study), Central and Southern Great Barrier Reef (two studies), Torres Strait Northern Great Barrier Reef (one study), Cape Howe (two studies), and Ningaloo (one study) (Figure 7b and Table 3).

Six studies were identified for the province of Cold Temperate Northwest Atlantic. The ecoregion included is Gulf of Maine/Bay of Fundy. Tropical Southwestern Atlantic and Warm Temperate Northwest Atlantic recorded four occurrences each, within the ecoregions of Eastern Brazil (two studies), North-eastern Brazil (two studies), Carolinian (three studies), and Northern Gulf of Mexico (one study). Tropical North-western Atlantic and Warm Temperate

Southwestern Atlantic each has two occurrences. The ecoregions under these two provinces, Floridian and South-eastern Brazil, respectively, have two studies each recorded.

The province of Somali/Arabian has three occurrences, the included ecoregion is Arabian (Persian) Gulf (three studies). Similarly, the province of Lusitanian also recorded three occurrences, in the ecoregion of Azores Canaries Madeira (two studies) and South European Atlantic Shelf (one study). Remaining recorded provinces are Western Indian Ocean (two occurrences) with the ecoregions of Mascarene Islands (one study) and Delagoa (one study); Agulhas (one occurrence) with the ecoregion Natal (one study), and South Kuroshio (one occurrence), with one study in the ecoregion South Kuroshio.

Within the included studies, majority (58 studies) of the study were conducted in regions of developed economies (United Nations Country Classification, accessed 2024), with Australia (23 studies) having the highest number of studies published within the category. This is followed by United States of America (12 studies) and Japan (seven studies). Within the category of developed economies, one study has also included an economy in transition (defined as industrialised countries that are undergoing process of transition to a market economy; these include some former Soviet Union republics [61]). Another study also encompasses a developed economy and developing economy (defined as countries with less developed industrial base and lower human development index relative to other countries [62]). For developing economies (31 studies), Hong Kong Special Administrative Region has the highest number of published studies (11 studies), followed by China (5 studies) and Brazil (5 studies).

For proximity of the research site to the mainland (Figure 8a), most studies involve close/territorial sea (69 studies). Middle/contiguous zone is represented in seven studies and far/exclusive economic zone in 15 studies. Of all the studies involving close/territorial sea, four studies also involve middle/contiguous zone, two studies involve far/exclusive economic zone, and two studies involve both middle/contiguous and far/exclusive economic zone. Ten studies

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428 did not specify their proximity to the mainland (Figure 8a) and 21 of the studies took place in
429 protected areas (Figure 8b). A list of the protected areas can be found in Table 4.

Methodological approaches.

Temperature Data Record. When examining the methods reported by the study authors for obtaining ocean temperature records, 50 publications were found to have obtained their temperature data *in-situ* (deployment of loggers), 28 studies used remote sensing data (herein referred to satellite derived and SST, sea surface temperature) for temperature data. 26 studies used other methods of obtaining temperature data, these are *ex-situ* conservation studies (where temperature was set to match the experimental criteria) (24 studies) or with study design not clearly identified (two studies) in the literature (Figure 9a). For *in-situ* collected data were collected using temperature logger (26 studies), dive computers (two studies), NOAA National Buoy Center (two studies), remotely operated vehicles (ROV) (two studies), digital thermometer (one study), mercury thermometer (one study), local monitoring sites (one study), and using T-MEDNet as a temperature data sharing platform (one study) (Figure 9b). 13 studies did not specify the method of recording temperature data. For temperature data availability, 51 of the 91 studies did not provide raw temperature data for access, with the remaining 40 studies providing full access to raw temperature data (see Supplementary Table S2).

Study depth. For study depth and/or depth of data collection, majority of temperature data were obtained at depth of 20 m below sea surface (Figure 9c), with one temperature data recording at a depth of 40 m (T-MEDNet). The highest occurrence of depth for temperature collection is 3 m (eight occurrences), followed by 4 m (seven occurrences) and 2 m (five occurrences).

Type and timing of study. 17 studies of the 91 studies included are reported to have taken place concurrently with extreme weather or climate events (Figure 10a and Table 5) and focused on reporting evidence associated with the specific event occurring, while two further studies report extreme weather events but were not conducted concurrently with the events. The majority of studies (72 studies) were not describing any specific extreme weather events.

Other environmental data. From the included studies, most observed environmental parameters (Figure 10b) are light (13 studies), salinity (12 studies) and pH values (nine studies), while other parameters include nutrients (eight studies), ocean current (six studies), chlorophyll a concentration (five studies), wind speed (one study), water quality (one study), mean-sea bottom temperature (one study), CO₂ concentration (one study), chlorophyll c concentration (one study), boron isotopic composition of sea water (one study), and bathymetry (one study).

Study species. For coral taxa studied, Scleractinian was the most dominant studied coral order (124 species studied), followed by Alcyonacea (seven species), Zoantharia (five species), Actiniaria (three species), Anthoathecata (two species), Axinellida (two species), Laminariales (two species), Agelasida (one species), Aplousobranchia (one species), Dictyoceratida (one species), Haplosclerida (one species), Pennatulacea (one species), Poecilosclerida (one species), and Suessiales (one species) (Figure 4a). Within the order Scleractinian, *Murilinidae* (48 occurrences) was found to be the most studied family of coral species, followed by Acroporidae (42 occurrences) (Figure 11). Species level information can be found in Supplementary Table S3,

Ocean Warming Evidence

69 studies provided an outcome (defined here as studies presenting results and conclusion based on their research efforts) associated with the study conclusion, while 13 studies provided prediction (defined here as studies providing a prediction of future scenarios based on results presented in their research) in their conclusion, and the remaining nine studies provided both outcome and prediction (Figure 12a). From the outcomes and predictions, most studies focus on reporting on physiology of coral systems under thermal stress (34 studies), followed by investigations into adaptation of coral species into marginal reefs (16 studies), changes in coral populations (12 studies), calcification of reefs (eight studies), assessment of potential refugia (six studies), recovery of population (five studies), climate change projections (two studies), impact of cold stress on coral (two studies), seasonal changes (two studies), and ENSO interactions (one study) (Figure 12b).

Drivers. Most studies considered long-term climate change as a driver of ocean warming (47 papers, 51.6%). This is followed by use of experimental stressors such as artificially raising temperature and acute thermal stress treatments (20 papers, 22.0%). Degree heating weeks were the third most recorded stressor indicator (11 papers, 12.1%) of ocean warming and marine heatwaves the fourth (eight papers, 8.8%). The other stressors include seasonal variation (four papers, 4.4%), localised hotspots (two papers, 2.2%), cold stress (two papers, 2.2%), El Nino Southern Oscillation (one paper, 1.1%), tidal temperature change (one paper, 1.1%), and temperature difference along latitudes (one paper, 1.1%). Two papers (2.2%) did not specify the stressor used when describing bleaching events (Figure 13a).

Severity. 17 papers (18.7%) reported degree heating weeks, marine heatwaves, or both as identifiers of the severity of ocean warming (Figure 13b), six papers did not specify the intensity of marine heatwaves, one paper recorded a moderate and a strong marine heatwave event, and one paper recorded a severe marine heatwave event. For studies reporting degree heating weeks as recording matrix, five papers did not specify the level of bleaching alert. Two separate papers recorded bleaching events exceeding bleaching alert level 5 (>20 DHW), another two papers recorded bleaching events exceeding bleaching alert level 4 (>16 DHW). Bleaching alert level 3 (>12 DHW) was reported in one paper, while bleaching alert level 2 (>8 DHW) was not reported in the papers. Three bleaching alert level 1 (>4 DHW) were reported in two separate papers, and three bleaching warnings (1 to 4 DHW) were reported in 1 paper (Figure 13b).

Limitation of the map

Limitations of the mapping process. While a comprehensive search strategy was developed and employed for this systematic map, it is essential to acknowledge the omission of relevant studies may have occurred, as the search string developed for this systematic map comprised of a combination of general and specific terms relevant to the topic. Our search string initially returned a large number of non-relevant literature, as we intended to capture as many relevant literatures as possible. Including more specific terms for future searches on database may assist in capturing a boarder spectrum of articles. Despite that, we believe our search string has successfully captured a board spectrum of articles available, enabling the mapping of evidence base [63].

In early 2010s, the number of publications identified investigating climate change impacts on subtropical reefs (or corals) was relatively low, ranging from 1 to 2 studies a year until 2014. A drastic increase in number of publications by 50% was then apparent from 2015 as research efforts begin to increase. An important review [37], highlighted the need for preserving high latitude coral reef systems and the potential impacts of climate change. Additionally, 2014 to 2017 saw an unprecedented period of severe sea surface temperature, and the longest recorded bleaching events for coral reef ecosystems which was associated high mortality in coral reefs [64]. Increasing awareness of, and observations of, climate change impact on subtropical coral reefs likely contributed significantly to the increase in research efforts into the subtropical reefs post 2014. The number of publications over the past 5 years continued showing a steady research effort being undertaken in the subtropical coral reefs, with now a near annual publication rate of 10 publications per year (2022, 2023).

Ecoregions, provinces and sources of climate change impacts evidence. Research efforts for subtropical coral research and subsequently evidence for the impact of climate

change in these ecosystems has predominantly focused on ecoregions in the Asian Pacific region. Specifically, the ecoregions of the provinces South China Sea, Andaman, Warm Temperate Northwest Pacific, and South Kuroshio are sites where research effort has been intensive (a total of 27 publications). Research in these sites were primarily published on and after 2015, associated with coral bleaching events reported in 2014 in Hong Kong [11], 2016 in Andaman Sea [65] and 2016 in Japan [13]. Therefore, ecoregions are also identified where there are research gaps and evidence for the impact of climate change has remained low, such as Delagoa, Mascarene Islands, Arabian Gulf, Natal, etc. While these ecoregions are directly exposed to the Western Indian Ocean, there is evidence of bleaching events in the area [66]. In this systematic map, studies where research effort has been associated with recorded marine protected areas were identified. Most marine park/reserve literature is reported from Australia. Only 23.1% (21 out of 91) of the studies reported having conducted research in protected or associated with marine protected areas and 35.2% of studies (32 out of 91) did not clearly indicate if their published studies took place in or within the region of a marine protected area. Given the recent increases in areas designated MPAs or marine parks internationally [67], future research identifying patterns of research effort, knowledge gaps and evidence synthesis for coral reef ecosystems should consider further investigating the sites within marine protected areas to build evidence-base for park management.

Drivers and Severity of Ocean Warming. Prior to 2016, most studies consider long-term climate change as an environmental driver for ocean warming. The use of long-term monitoring data was common for studies, and the research using this as an approach to study drivers of ocean warming generally investigate coral calcification and adaptation to marginal reefs to examine the effect of ocean warming on coral health and habitats. The introduction of a hierarchical approach developed to defining marine heatwaves through 30 years of temperature record of the regions [32] led to the increase in use of marine heatwaves as a tool for identifying potential bleaching events [68]. Use of degree heating weeks for subtropical

corals was uncommon until 2017, when an increased number of publications used degree heating week to quantify thermal stress of subtropical coral [60, 69, 70] was observed. This allowed the authors of respective studies to use standardised metrics to compare the risk of bleaching on subtropical reefs to tropical reefs. The use of marine heatwave and degree heating weeks provide researchers with quantitative metrics for evaluating severity of an ocean warming event and/or bleaching event. A combined use of both marine heatwave and degree heating week as reporting mechanisms may benefit future studies in quantitatively comparing data between sites and events.

Implications for research. Research globally has primarily focused on reporting ecological responses to climate change associated events; however, substantial effort has also been directed to experimental studies. Physiological data is one of the most common forms of impact measures, and data was reported in the studies included in our map, with 30% of the identified literature reporting on the physiological state of corals associated with climate impacts in subtropical regions. Future research in this area should consider a meta-analysis of the evidence identified in the current study. In particular, there were severe bleaching events recorded from 2010-2023 occurring within the mortality scale of impact (degree heating week of 8 or above). Collating the thermal stress reports across subtropical regions will aid managers in predicting and responding to the impact of bleaching on these reefs. Ongoing systematic assessment of research effort and evidence will also allow for the evidence-based management practices specific to these unique and valuable ecosystems.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The datasets generated and/or analysed during the current study are available in the GitHub repository, https://github.com/MLH95/Ho-et-al-2024_Systematic-Map

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

Publication Authors: Ho, M.L.¹, Leggat WP², Gaston T, Nakagawa, S Perkins-Kirkpatrick S, Hobday AJ, Richards Z Page C.E.¹, Sawyers P, Lagisz, M & Ainsworth T.D².

MLH and TDA conceptualised the idea for this study. MLH, TDA, ML and SN designed the search strategy. MLH, TDA, PS and CP designed the data extraction methods, as well as validating the design. MLH and PS screened the original database of articles by title and abstract, and later screening was done collaboratively by MLH, PS, CP, SV, JW and SE. Data analysis was performed by MLH. MLH led the writing of the report with crucial contribution from TDA. All authors edited the manuscript draft and give their permission for publication of this systematic map in its final form.

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Tables

Table 1 List of Authors and the number of publications they are involved in.

Author Name	Number of Publication Involved
Ang, Put	7
Ross, Claire	5
Chui, Apple Pui Yi	4
Huang, Xueyong	4
Linares, Cristina	4
McCulloch, Malcolm	4
Qiu, Jian-Wen	4
Tsang, Ryan Ho Leung	4
Yu, Kefu	4
Agostini, Sylvain	3
Beger, Maria	3
Byrne, Maria	3
Cebrian, Emma	3
Chen, Chaolun Allen	3
Dalton, Steven	3
Davies, Sarah	3
Davy, Simon	3
Denis, Vianney	3
Ferrier-Pages, Christine	3
Figueira, Will	3
Garrabou, Joaquim	3
Goffredo, Stefano	3
Leggat, William	3
Qin, Zhenjun	3
Schoepf, Verena	3
Sommer, Brigitte	3
Aichelman, Hannah	2
Ainsworth, Tracy	2
Airi, Valentina	2
Barrett, Neville	2
Brito, Alberto	2
Cant, James	2
Carroll, Andrew	2
Cerrano, Carlo	2
Chan, Leo	2
Chen, Biao	2
Clemente, Sabrina	2
Day, Paul	2
De Palmas, Stephane	2
Dellisanti, Walter	2

Dove, Sophie	2
Dubinsky, Zvy	2
Falini, Giuseppe	2
Fisher, Paul	2
French, Ben	2
Hoegh-Guldberg, Ove	2
Hovey, Renae	2
Huang, Wen	2
Hwang, Sung-Jin Jin	2
Krueger, Thomas	2
Kruži?, Peter	2
Liang, Jiayuan	2
López, Cataixa	2
López-Sendino, Paula	2
Loubeyres, Mathilde	2
Malcolm, Hamish	2
Marchini, Chiara	2
McIlroy, Shelby	2
Nakamura, Takashi	2
Pandolfi, John	2
Pontasch, Stefanie	2
Qian, Pei-Yuan	2
Reimer, James	2
Ribas-Deulofeu, Lauriane	2
Rotjan, Randi	2
Sampayo, Eugenia	2
Song, Jun-Im	2
Suggett, David	2
Teixidó, Núria	2
Wang, Guanghua	2
Wells, Mark	2
Woo, Seonock	2
Wu, Jiajun	2
Xie, James	2
Yamano, Hiroya	2
Yeung, Yip Hung	2
Aizawa, Hiroaki	1
Alfaro-Lucas, Joan	1
Almanzar, Adeline	1
Angeletti, Lorenzo	1
Aurelle, Didier	1
Baird, Andrew	1
Baird, Maria	1
Baker, David	1

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2	Baldi, Veronica	1
3	Bally, Marc	1
4	Banha, Thomás	1
5	Barbosa, Andreia	1
6	Barshis, Daniel	1
7	Bas-Silvestre, Maria	1
8	Bates, Amanda	1
9	Bauman, Andrew	1
10	Baumann, Justin	1
11	Beal, Jeff	1
12	Beals, Morgan	1
13	Bednarek, Teresa	1
14	Bencons, Brooke	1
15	Bensoussan, Nathaniel	1
16	Béraud, Eric	1
17	Bergman, Jessica	1
18	Besemer, Nicole	1
19	Bessell-Browne, Pia	1
20	Bonebrake, Timothy	1
21	Booth, David	1
22	Braga, Marcus	1
23	Brandt, Angelika	1
24	Brennan, Sara	1
25	Bridge, Tom	1
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27	Bryson, Mitch	1
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39	Castillo, Karl	1
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41	Chang, Taison Ka Tai	1
42	Chaudhary, Chhaya	1
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Chen, Zesheng	1
Choi, Kwang-Sik	1
Chow, Wing-Kuen	1
Chung, Jeffrey	1
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Clode, Peta	1
Cocito, Silvia	1
Cohen, Anne	1
Coma, Rafel	1
Comeau, Steeve	1
Cook, Katie	1
Cooper, Antonia	1
Coppari, Martina	1
Cornwall, Christopher	1
Correa, Rogger	1
Courtney, Travis	1
Coutinho, Ricardo	1
Crisci, Caroline	1
Dafforn, Katherine	1
Davis, Kay	1
Davis, Tom	1
de Bettignies, Thibaut	1
de Caralt, Sònia	1
De, Kalyan	1
DeCarlo, Thomas	1
Dellaert, Zoe	1
Denis-Roy, Lara	1
Di Camillo, Cristina	1
Diamond, Sandra	1
Dickerson, Hayden	1
Dimond, James	1
Ding, Wei	1
Diniz, Mário	1
Dunic, Jillian	1
Eakin, Mark	1
Edgar, Graham	1
Edgar, Robert	1
Emslie, Michael	1
Enochs, Ian	1
Entwistle, Kristina	1
Escolà, Marta	1
Faleiro, Filipa	1
Falter, James	1

Falter, Jim	1
Féral, Jean-Pierre	1
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Ferreira, Lucas Cabral Lage	1
Figueiredo, Joana	1
Filho, Fernando	1
Finnerty, John	1
Floc'h, Nicolas	1
Flores, Augusto	1
Foster, Taryn	1
Francini, Carlo	1
Francini-Filho, Ronaldo	1
Fraser, Nicole	1
Frieta-Vali?, Maša	1
Fromont, Jane	1
Fuchs, Yann	1
Fujise, Lisa	1
Fumani, Neda	1
Gevaert, Francois	1
Gili, Josep-Maria	1
Gilliam, David	1
Giraldo-Ospina, Ana	1
Gizzi, Francesca	1
Glassom, David	1
Gomez-Cabrera, Maria	1
Gómez-Gras, Daniel	1
Gong, Sanqiang	1
González-Pech, Raúl A.	1
Gori, Andrea	1
Goyen, Samantha	1
Guidi, Elena	1
Gurgel, Anne	1
Hadjioannou, Louis	1
Hall-Spencer, Jason	1
Harborne, Alastair	1
Harrison, Peter	1
Harvey, Ben	1
Hashtroudi, Mehri	1
Hay, Mark	1
Heather, Freddie	1
Heron, Scott	1
Hicks, Jamie	1
Higuchi, Tomihiko	1
Hill, Ross	1

Hodge, Jessica	1
Holcomb, Michael	1
Howe, Steffan	1
Hsieh, Hernyi	1
Huang, Hui	1
Huang, Rongyong	1
Huang, Zhi	1
Huges, Terence	1
Hughes, David	1
Iguchi, Akira	1
Ingole, Baban	1
Ip, Jack Chi-Ho	1
Jami, Mohammad Javad	1
Janquin, M-A	1
Jiang, Leilei	1
Jimenez, Carlos	1
Jin, Xuejie	1
Johnson, Olivia	1
Johnston, Emma	1
Johnston, Nicole	1
Jones, Nicolas	1
Jones, Tyson	1
Jordan, Alan	1
Júnior, Francisco	1
Kahike, Tim	1
Kaufman, Les	1
Kavousi, Javid	1
Kei, Keith	1
Keith, Sally	1
Kendrick, Gary	1
Kenkel, Carly	1
Kersting, Diego	1
Kerwin, Allison	1
Keshavmurthy, Shashank	1
Kim, Sun	1
Kirkland, Amanda	1
Kitahara, Marcelo	1
Kitano, Yuko	1
Klepac, Courtney	1
Knott, Nathan	1
Kolodziej, Graham	1
Komatsu, Kosei	1
Kon, Koetsu	1
Kurihara, Haruko	1

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2	Kyriacou, Nicole	1
3	La Rivière, Marie	1
4	La Riviere, Patrick	1
5	Lachs, Liam	1
6	LaJeunensse, Todd	1
7	Larkin, Meryl	1
8	Lau, Dickey	1
9	Le Nohaïc, Morane	1
10	Ledoux, Jean-Baptiste	1
11	Legorreta, Renata	1
12	Lester, Emily	1
13	Leung, Yu Hin	1
14	Levy, Oren	1
15	Lewis, Brett	1
16	Li, Ming	1
17	Li, Shu	1
18	Liao, Zhuiheng	1
19	Liesegang, Mary	1
20	Lima, Laís	1
21	Ling, Scott	1
22	Lipej, Lovrenc	1
23	Liu, Gang	1
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25	Lloyd, Alicia	1
26	Longo, Guilherme Ortigara	1
27	Lopes, Ana	1
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29	Lucas, Caroline	1
30	Luo, Yanqiu	1
31	Mantas, Torcuato	1
32	Manzello, Derek	1
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36	Matthews, Jennifer	1
37	Matz, Mikhail	1
38	Mavri?, Borut	1
39	McClanahan, Tim	1
40	McCorkle, Daniel	1
41	McMahon, Ashly	1
42	Mellin, Camille	1
43	Meng, Linqing	1
44	Menu, Dominique	1

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5	Mo, Shaohua	1
6	Mokhtar-Jamaï, Kenza	1
7	Montagna, Paolo	1
8	Monteiro, Ana	1
9	Mooney, Peter	1
10	Moreno, Sergio	1
11	Moriarty, Tess	1
12	Mote, Sambhaji	1
13	Muthiga, Nyawira	1
14	Nagai, Satoshi	1
15	Nakabayashi, Aki	1
16	Nakamura, Masako	1
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18	Negaresta, Hossein	1
19	Ng, Ysz Yan	1
20	Nitschke, Matthew	1
21	Nomura, Keiichi	1
22	Nozawa, Yoko	1
23	Oakley, Clinton	1
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25	Oh, Elizabeth	1
26	Orejas, Covadonga	1
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28	Paiva, Sandra	1
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30	Peirano, Andrea	1
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32	Pizarro, Oscar	1
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39	Ralph, Peter	1
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43	Reyes, Christopher	1
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Reynaud, Stéphanie	1
Ribes, Marta	1
Ricci, Francesco	1
Rivera, Hanny	1
Roberson, Loretta	1
Robert, Keyzers	1
Rodic, Petra	1
Rodríguez, Adriana	1
Roff, George	1
Rosa, Inês	1
Rosa, Rui	1
Rottier, Cecile	1
Roveta, Camilla	1
Saeedi, Hanieh	1
Saleh, Abolfazl	1
Salguero-Gómez, Roberto	1
Salih, Anya	1
Samaroo, Jason	1
Samiei, Jahangir Vajed	1
Santos, Isaac	1
Schleyer, Michael	1
Schmidt-Roach, Sebastian	1
Schniedewind, Mikarla	1
Schutter, Miriam	1
Seroussi, Yanir	1
Serra, Ignasi	1
Sfenthourakis, Spyros	1
Sharp, Koty	1
Sharp, Victoria	1
Shields, Derek	1
Shields, Joe	1
Shirvani, Arash	1
Short, JA	1
Simões, Marianna	1
Sims, Carrie	1
Sink, Kerry	1
Skirving, William	1
Smit, Kaylee	1
Smith, Stephen	1
Soares, Marcelo	1
Soler, German	1
Souza, Fabio	1
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Stat, Michael	1

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2	Stewart, FJ	1
3	Strudwick, Paige	1
4	Stuart-Smith, Jemina	1
5	Stuart-Smith, Rick	1
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7	Sumida, Paulo	1
8	Sun, Jin	1
9	Suzuki, Yoshimi	1
10	Sweatman, Hugh	1
11	Tan, Yehui	1
12	Taviani, Marco	1
13	Taylor, Michael	1
14	Teixeira, Carlos	1
15	Teshima, Kosuke	1
16	Thompson, Philip	1
17	Thomson, Damian	1
18	Thornhill, Daniel	1
19	Tian, Renmao	1
20	Tong, Haoya	1
21	Torsani, Fabrizio	1
22	Tortorelli, Giada	1
23	Townsend, Joseph	1
24	Tramonte, Carlos	1
25	Trubenbach, Katja	1
26	Trumble, Isabela	1
27	Tuckett, Chenae	1
28	Turak, Emre	1
29	Turnbull, John	1
30	Valisano, Laura	1
31	Vargas, Phillip	1
32	Viladrich, Núria	1
33	Vinagre, Catarina	1
34	Voss, Joshua	1
35	Wada, Shigeki	1
36	Wang, Wenhuan	1
37	Wang, Yinghui	1
38	Wang, Yonggang	1
39	Wernberg, Thomas	1
40	Wicks, Laura	1
41	Wilkinson, Shaun	1
42	Williams, Stefan	1
43	Willians, Maureen	1
44	Wilson, Shaun	1

Wong, Tim	1
Woolsey, Erika	1
Wu, Qian	1
Wuitchik, Daniel	1
Xia, Xiaomin	1
Xu, Shendong	1
Yamakita, Takehisa	1
Yang, Enguang	1
Yasuda, Nina	1
Yeung, Chung Wing	1
Yiu, Sam	1
Yu, Vriko	1
Yu, Wanjun	1
Yuan, Felix	1
Yuyama, Ikuko	1
Zhang, Weipang	1
Zhang, Wenjing	1
Zhang, Yanjie	1
Zhou, Guowei	1
Zimmerman, Richard	1
Zuberer, Frederic	1

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808 Table 2 Number of publications by journals across literature included for this systematic map.

Journal Name	Number of Publications
Coral Reefs	17
Frontiers in Marine Science	7
Scientific Reports	6
Science of The Total Environment	4
Diversity	3
Global Change Biology	3
Marine Biology	3
Marine Ecology Progress Series	3
Marine Pollution Bulletin	3
PloS One	3
Proceedings of the Royal Society B	3
Bulletin of Marine Science	2
Ecology and Evolution	2
Journal of Experimental Marine Biology and Ecology	2
Limnology and oceanography	2
Molecular Ecology	2
Applied microbiology and biotechnology	1
Aquatic Conservation: Marine and Freshwater Ecosystems	1
Aquatic Ecosystem Health & Management	1
Biodiversity and Conservation	1
Biogeosciences	1
Cell Stress and Chaperones	1
Ecography	1
Environmental Science and Pollution Research	1
Estuarine, Coastal and Shelf Science	1
Frontiers in Physiology	1
Geophysical Research Letters	1
Helgoland Marine Research	1
Integrative Organismal Biology	1
Journal of Experimental Biology	1
Journal of Heredity	1
Journal of Marine Science and Engineering	1
Journal of phycology	1
Marine Chemistry	1
Marine Environmental Research	1
Microorganisms	1
Nature	1
Pakistan Journal of Biological Sciences	1
PeerJ	1
Progress in Oceanography	1
Regional Studies in Marine Science	1
The Biological Bulletin	1

810 Table 3 Number of publications by ecoregions across this systematic map.

Ecoregions	Belongs to Province	Number of Publications Concerning the Ecoregion
Southern China	South China Sea	15
Western Mediterranean	Mediterranean Sea	11
Lord Howe and Norfolk Islands	Lord Howe and Norfolk Islands	9
Leeuwin	Southwest Australian Shelf	6
Gulf of Maine/Bay of Fundy	Cold Temperate Northwest Atlantic	6
Central Kuroshio Current	Warm Temperate Northwest Pacific	6
East China Sea	Warm Temperate Northwest Pacific	6
Houtman	West Central Australian Shelf	5
Gulf of Tonkin	South China Sea	4
Arabian (Persian) Gulf	Somali/Arabian	3
Tweed-Moreton	East Central Australian Shelf	3
Carolinian	Warm Temperate Northwest Atlantic	3
Central and Southern Great Barrier Reef	Northeast Australian Shelf	2
Azores Canaries Madeira	Lusitanian	2
Southeastern Brazil	Warm Temperate Southwestern Atlantic	2
Floridian	Tropical Northwestern Atlantic	2
Eastern Brazil	Tropical Southwestern Atlantic	2
Cape Howe	Southeast Australian Shelf	2
Northeastern Brazil	Tropical Southwestern Atlantic	2
Adriatic Sea	Mediterranean Sea	2
Torres Strait Northern Great Barrier Reef	Northeast Australian Shelf	1
Northern Gulf of Mexico	Warm Temperate Northwest Atlantic	1
Levantine Sea	Mediterranean Sea	1
Natal	Agulhas	1
Western India	West and South Indian Shelf	1

West Caroline Islands	Tropical Northwest Pacific	1
Andaman Sea Coral Coast	Andaman	1
South Kuroshio	South Kuroshio	1
Ningaloo	Northwest Australian Shelf	1
Ionian Sea	Mediterranean Sea	1
Mascarene Islands	Western Indian Ocean	1
Manning-Hawkesbury	East Central Australian Shelf	1
South China Sea Oceanic Islands	South China Sea	1
South European Atlantic Shelf	Lusitanian	1
Delagoa	Western Indian Ocean	1

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812 Table 4 List of identified marine reserve, protected area, or marine park.

Name of Protected Area	Country	Reference
Abrolhos Marine National Park	Brazil	Ferreira et al., 2021
Blue Bay Marine Park	Mauritius	McClanahan et al., 2021
Columbretes Islands Marine Protected Area	Spain	Kersting et al., 2015
Medes Island Marine Protected Area	Spain	Kersting et al., 2015; Gómez-Gras et al., 2019; Crisci et al., 2017
Green Island Marine Research Station	Taiwan	Schutter et al., 2015
Jurien Bay Marine Park	Australia	Ross et al., 2021
Marmion Marine Park	Australia	Ross et al., 2021
Rottneest Island Marine Reserve	Australia	Ross et al., 2021
Shoalwater Islands Marine Park	Australia	Ross et al., 2021
Ngari Capes Marine Park	Australia	Ross et al., 2021
Lord Howe Island Marine Park	Australia	Steinberg et al., 2022; Davis et al., 2020; Oakley et al., 2022; Dalton et al., 2011
Malvan Marine Sanctuary	India	De et al., 2023
Scandola Nature Reserve	France	Crisci et al., 2017
Mljet National Park	Croatia	Kružić et al., 2015
Pedra da Risca do Meio	Brazil	Lucas et al., 2023
Pedra Vermelha Restricted-Access Site	Brazil	Lima et al., 2016
Solitary Islands Marine Park	Australia	Lachs et al., 2021; Dalton et al., 2011
Tung Ping Chau Marine Park	Hong Kong	Chui et al., 2015; Chui et al., 2016; Chui et al., 2017; Chui et al., 2023
Yan Chau Tong Marine Park	Hong Kong	Cai et al., 2018

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814 Table 5 Studies that were conducted concurrently with extreme weather events.

Period of Extreme Weather Events	Study
Beginning of Austral Summer 2010 to End of Austral Summer 2011	Bridge et al., 2014
January 2019 to March 2019	Davis et al., 2020
2015 to 2016	Le Nohaïc et al., 2017
February 2016 to August 2016	Goyen et al., 2019
2020 to 2021	Mo et al., 2022
March 2019 to May 2019	Ferreira et al., 2021
Multiple events over the years of 2004 to 2019	McClanahan et al., 2021
February 2019 to May 2019	Banha et al., 2020
June 2017 to August 2017	Ip et al., 2022
August 2014 to September 2014	Xie et al., 2017
Within 2020	Bergman et al., 2023
February 2011 to April 2011	Giraldo-Ospina et al., 2020
December 2018 to January 2019, January 2019 to February 2019, March 2019 to June 2019	Steinberg et al., 2022
1997 to 1999, 2001 to 2004, 2006 to 2009, 2011 to 2012	Kružić et al., 2014
April 2016 to July 2016	Lachs et al., 2021
2014 to 2017	Porter et al., 2021
January 2020 to February 2020, February 2020 to May 2020, May 2020 to June 2020	Lucas et al., 2023

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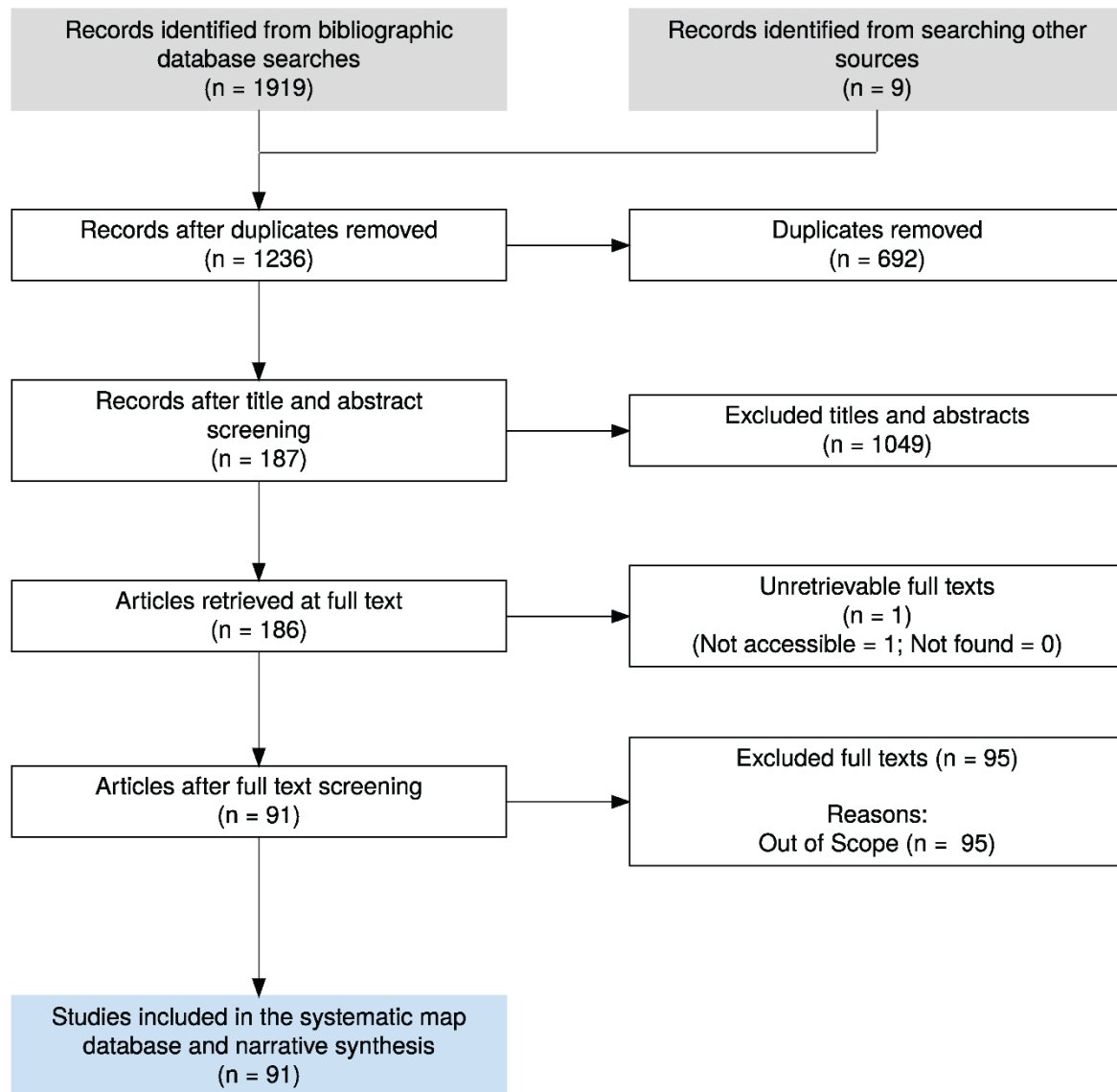


Figure 1 Flow diagram of this systematic map, detailing the process of removal literature per stage. Created with ROSES reporting flow diagram tool.



Figure 1 a) Word cloud featuring the most frequent keyword used for the literature included in this systematic map. Created using R package wordcloud. b) The number of publications by year, created using R package ggplot2.



Figure 3 Map showing tropical ecoregions (Yellow), subtropical ecoregions (Green), temperate ecoregions (Blue) and arctic ecoregions (Light Blue), created using R package leaflet.

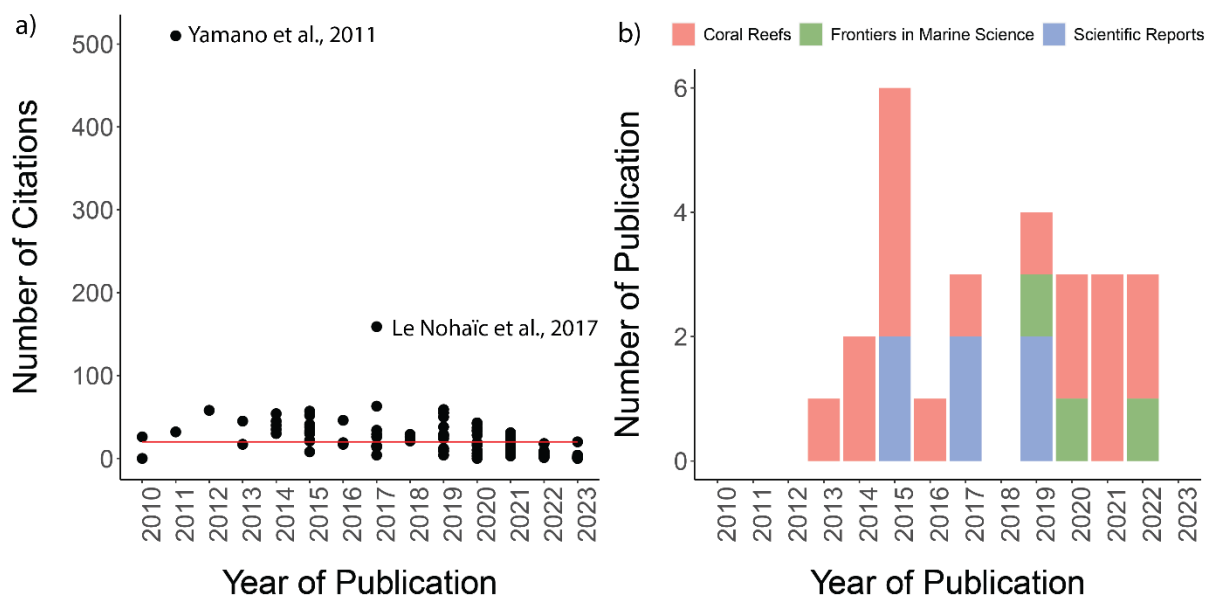


Figure 4a) Number of citations of literature by years. Red line is the median of number of citations across the literature included in this systematic map. b) The top 3 journals by number of publications by year. Created using R package ggplot2.

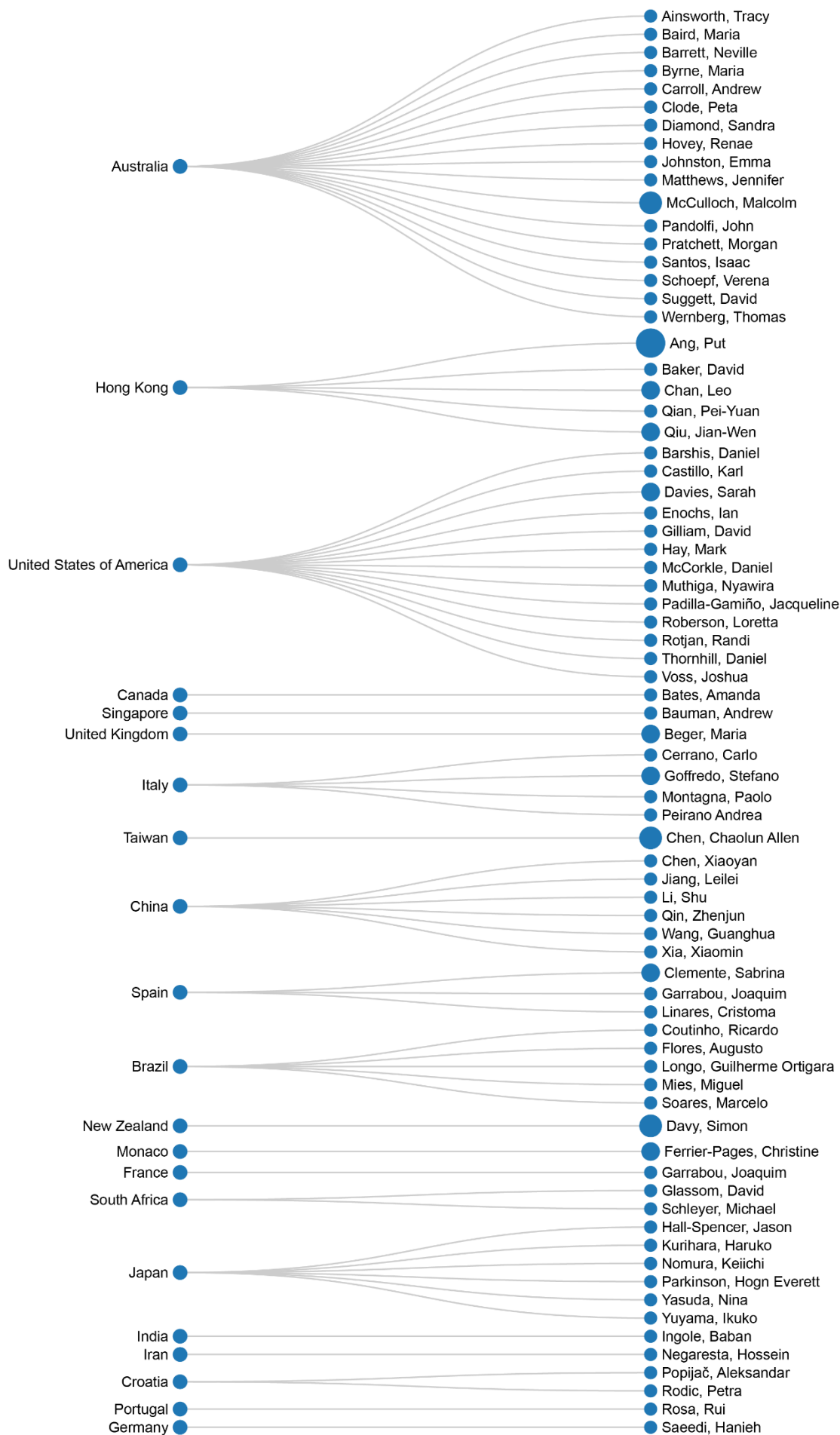


Figure 5 Linear dendrogram showing relations of last authors to countries their institutions belong to, created using RawGraphs.io.

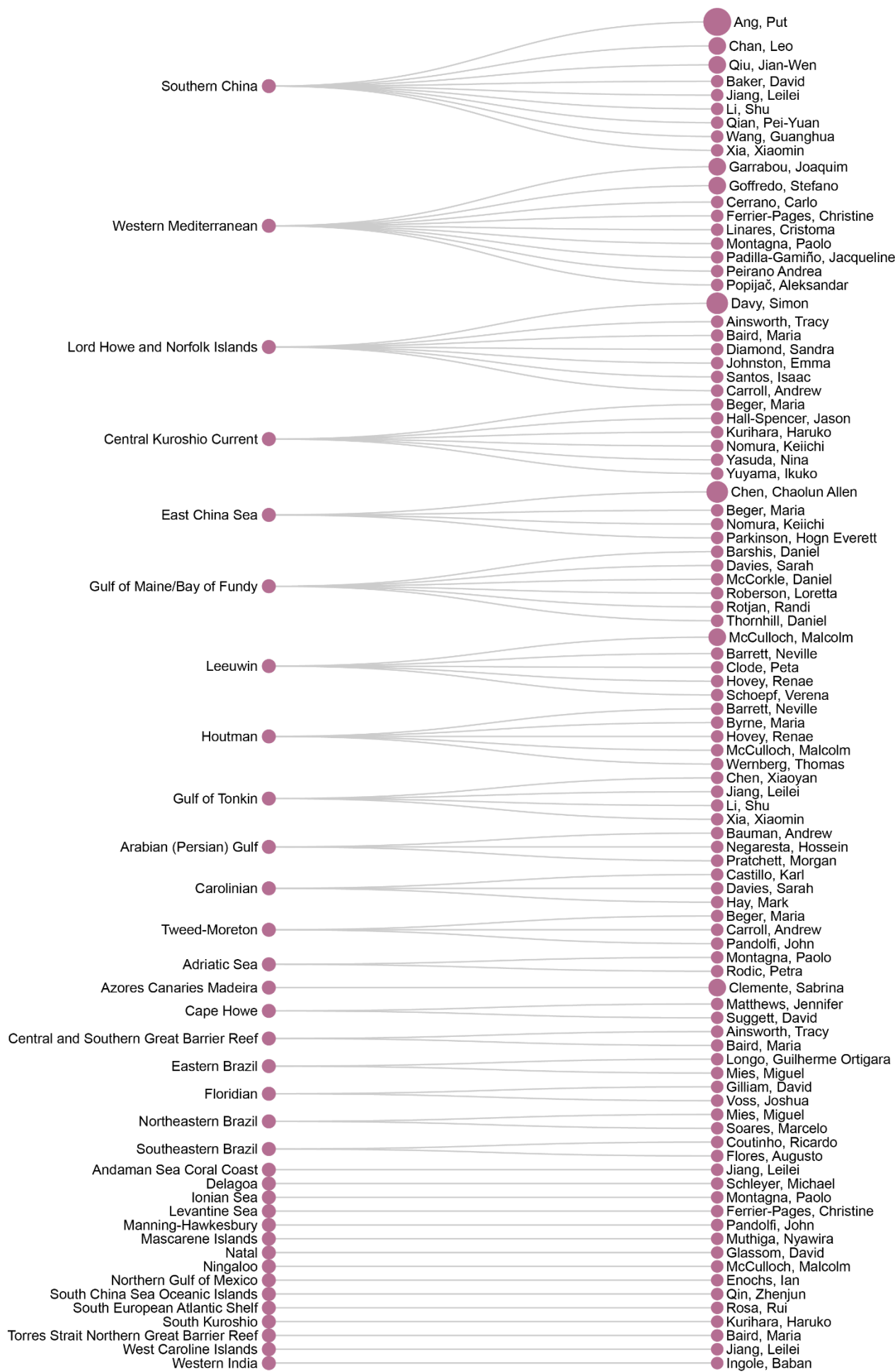


Figure 6 Linear dendrogram showing relations of last authors to ecoregions their literature focused on, created using RawGraphs.io.

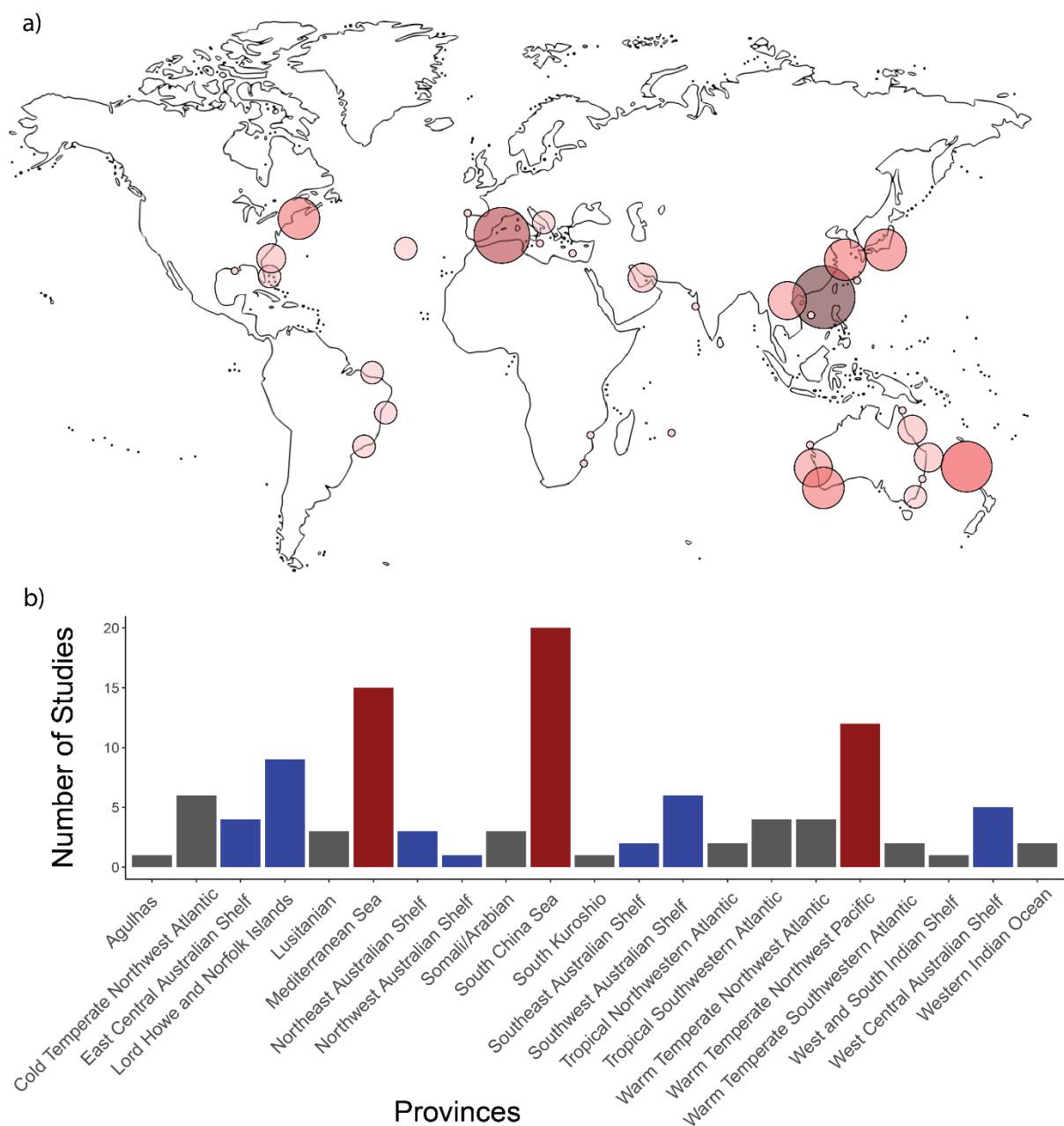


Figure 7a) World map showing the knowledge cluster by ecoregion, created using R package ggplot2. b) Number of studies by provinces, showing top 3 provinces (Red) and provinces in Australia (Blue), created using R package ggplot2.

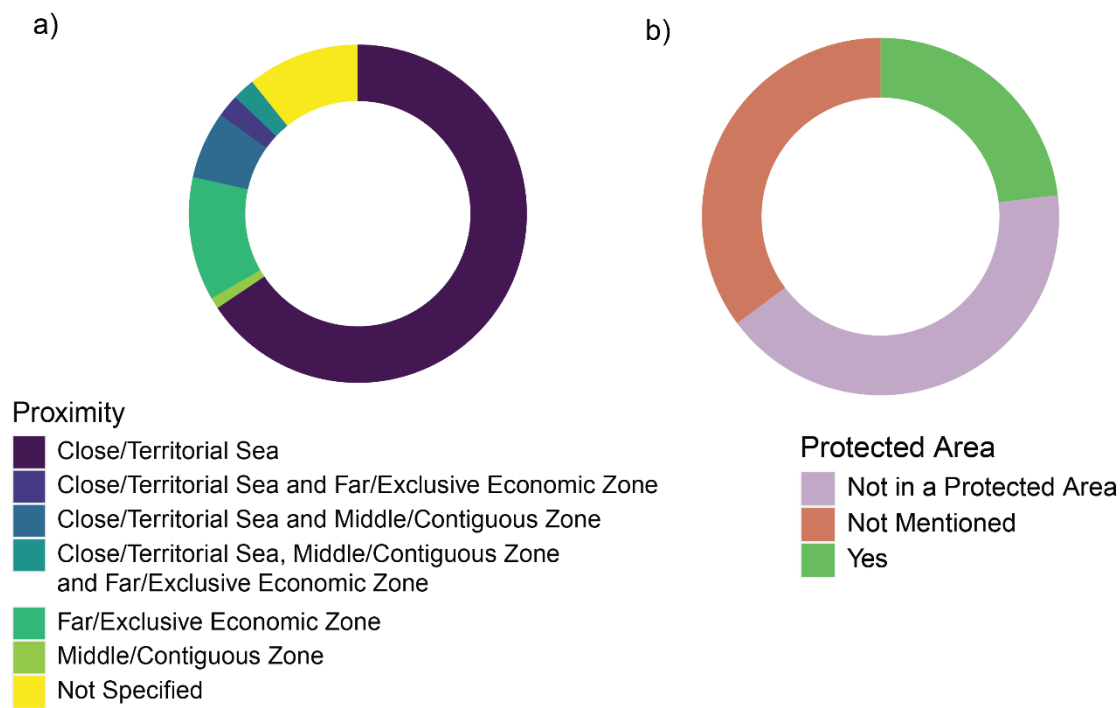


Figure 8a) Donut plot showing the proximity of the locations of studies to the closest EEZ across all literature in this systematic map, created using R package ggplot2. b) Donut plot showing the proportions of studies that took place in a protected area, created using R package ggplot2.

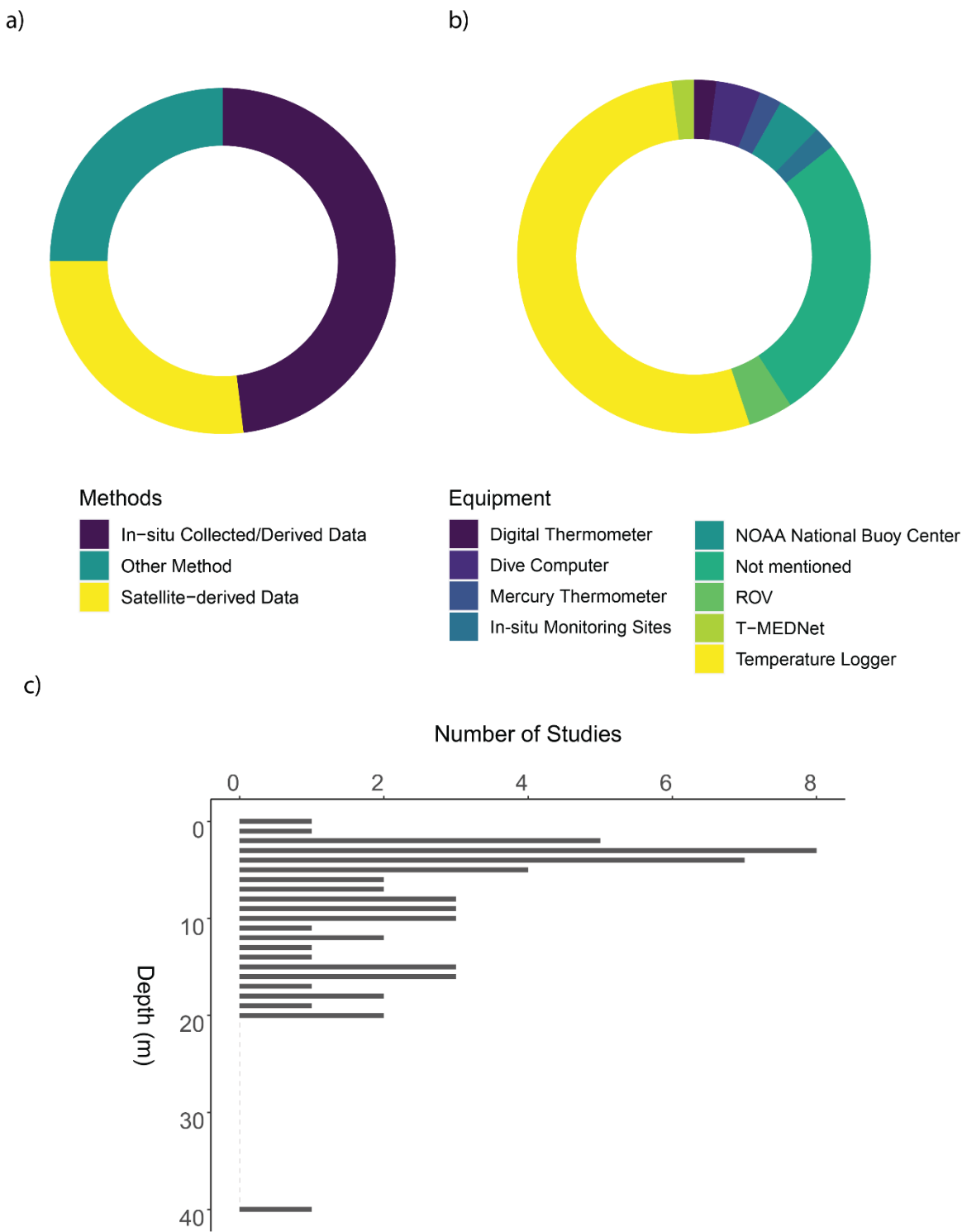


Figure 9 a) Proportion of methods of temperature recording used across the literature included in this study, created using R package ggplot2. b) Donut chart showing the proportion of equipment used for studies that recorded temperature in-situ, created using R package ggplot2. c) The depth of temperature data collection, created using R package ggplot2.

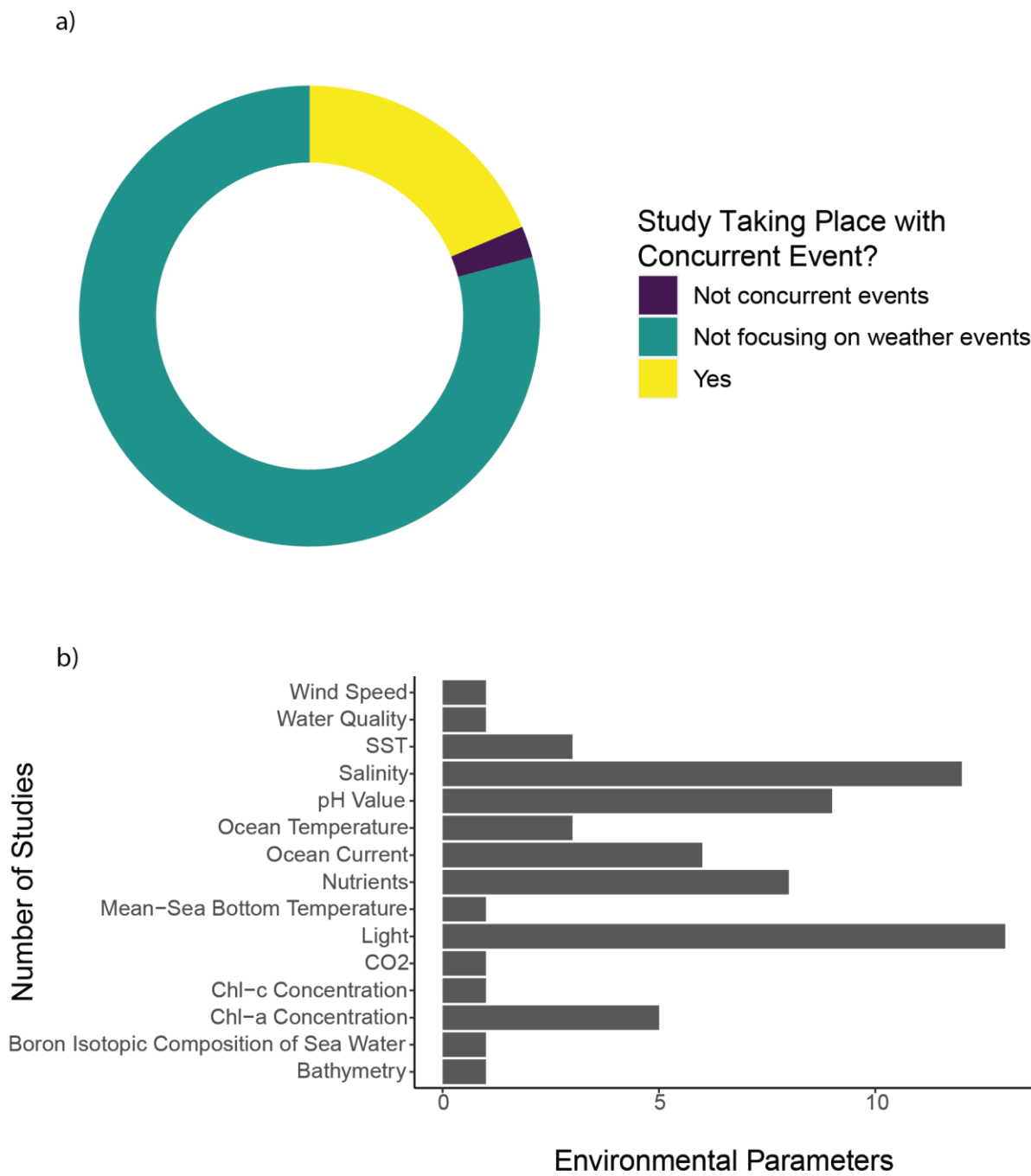


Figure 10a) Portion of publication focusing on weather event, created using R package ggplot2. b) Other parameters recorded in the literature covered in this systematic map, created using R package ggplot2.

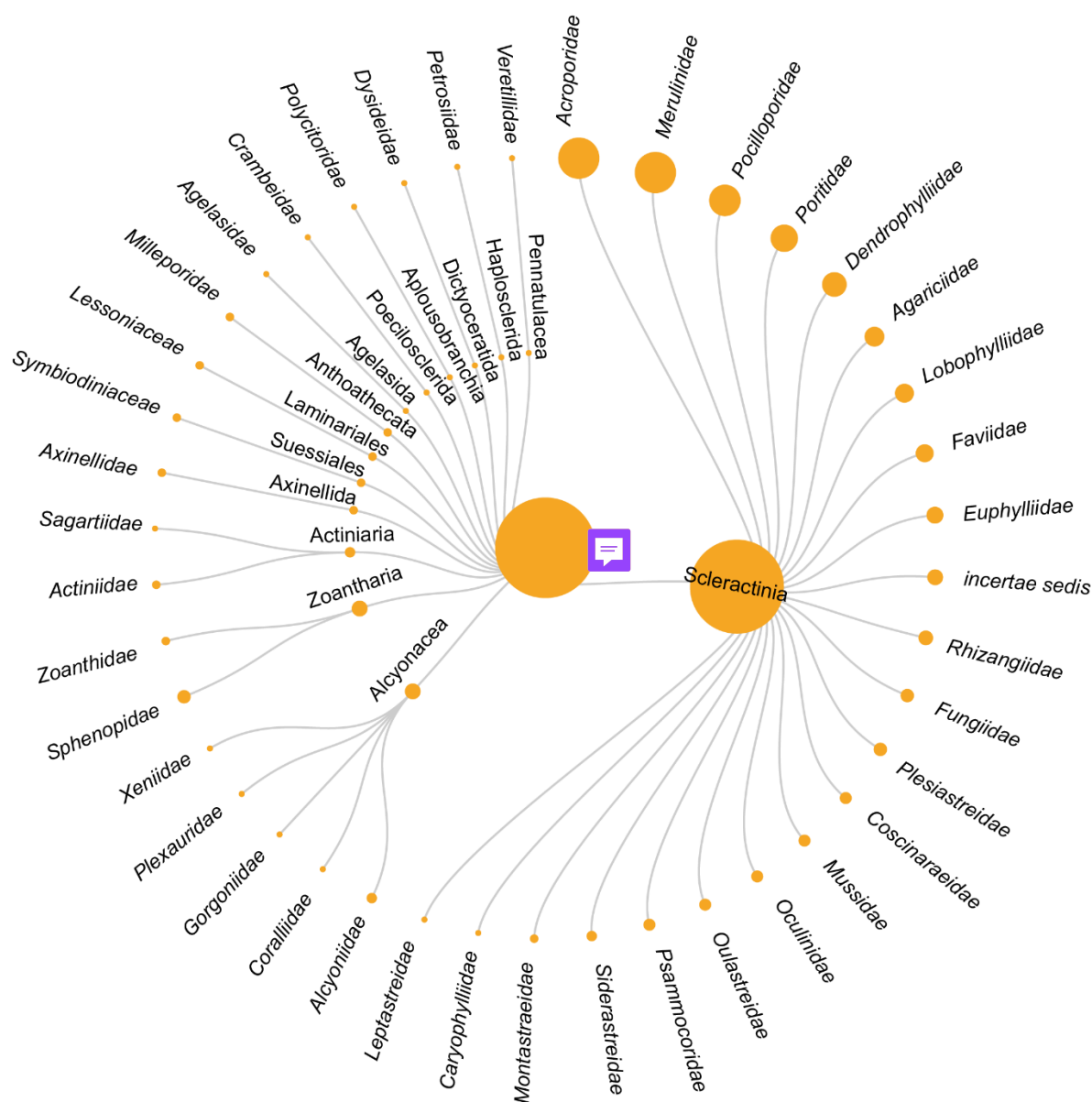


Figure 11 Dendrogram showing the families studied by the literature recorded in this systematic map.

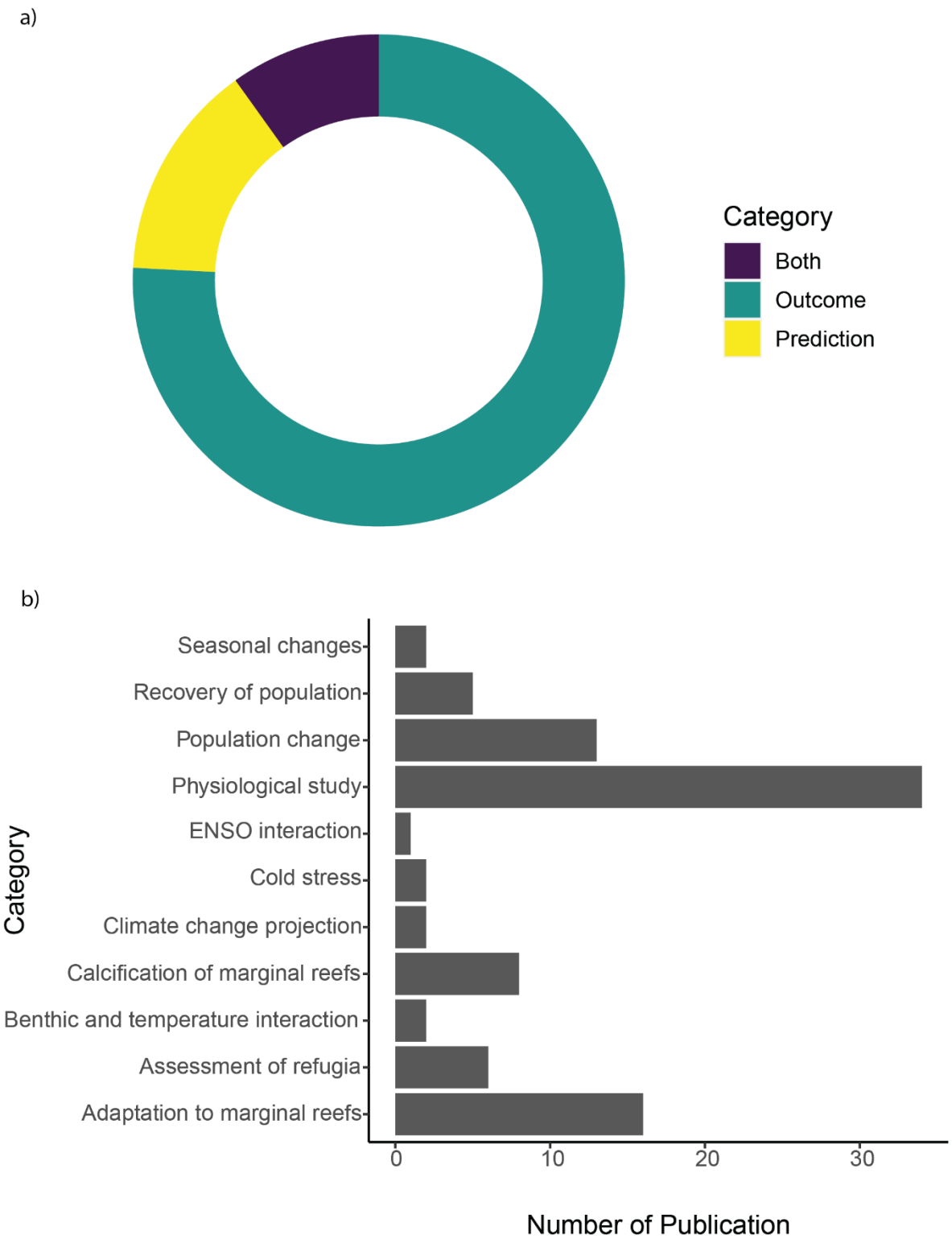


Figure 12 Proportion of literature included in this systematic map that include outcome and/or prediction, created using R package ggplot2. b) Categories of outcome/prediction and their number of publications respectively, created using R package ggplot2.

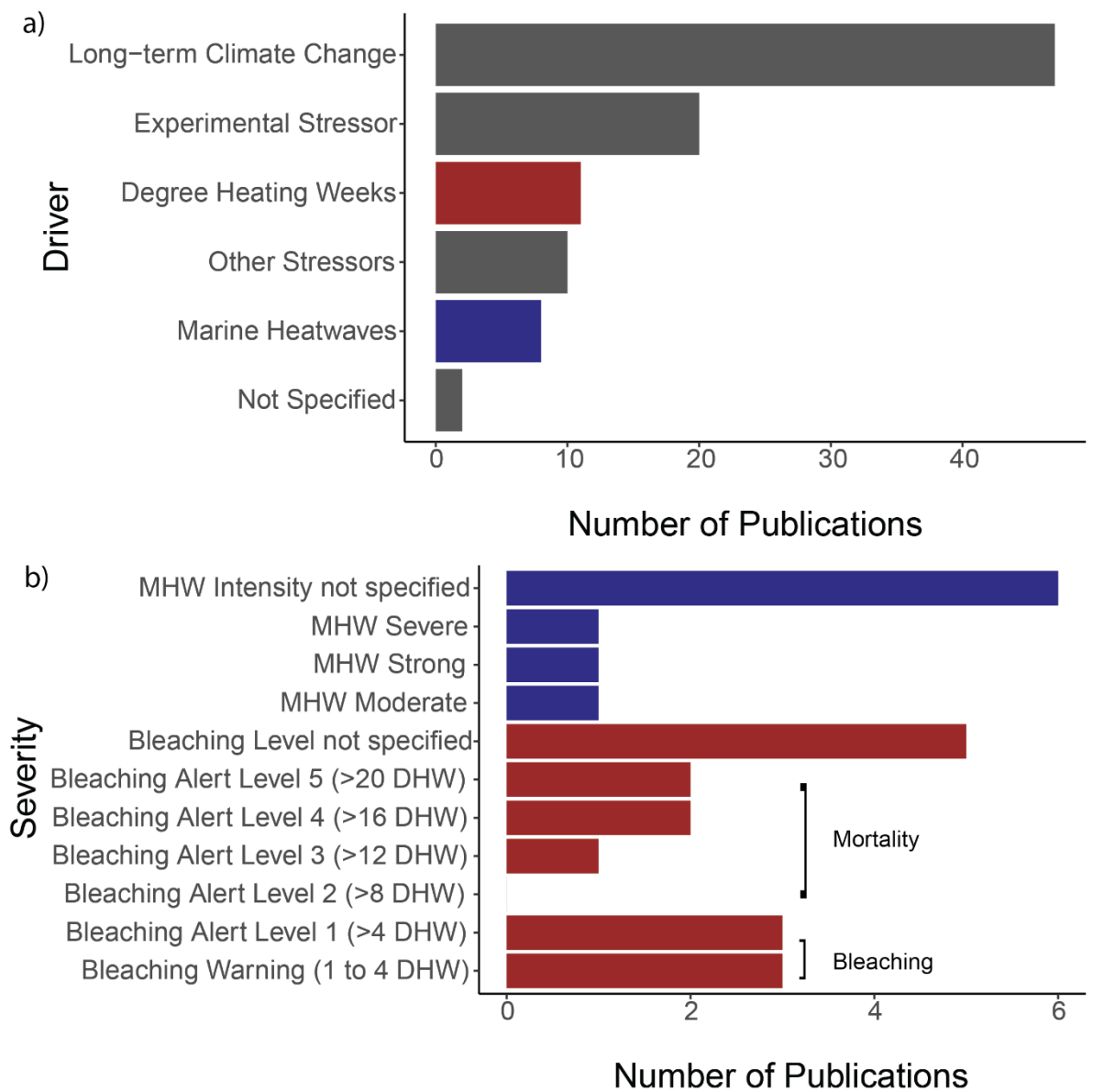


Figure 13 a) Drivers of extreme weather events recorded, including degree heating weeks (DHW) (Red) and marine heatwaves (MHW) (Blue), created using R package ggplot2. b) Number of publications reporting the severity of DHW bleaching alerts and MHW severity (Hobday et al., 2016), created using R package ggplot2.









Item number	Section/sub-section	Topic	Description	Further explanation	Checklist/meta-data	Author response	Comments
1	Title	Title	The title must indicate that it is a systematic map, and should indicate if it	The title should normally be the same or very similar to the review	Meta-data	What is the evidence for the impact of ocean warming on subtropical and temperate corals and coral reefs? A systematic map.	
2	Type of review	Type of review	Select one of the following types of review: systematic map, systematic	See CEE Guidance on systematic mapping [1], and on amendments and	Meta-data	systematic map	
3	Authors' contacts	Authors' contacts	The full names, institutional addresses and email addresses for all		Checklist	Yes	
4	Abstract	Structured summary	The abstract of the manuscript must not exceed 500 words and must be		Checklist	Yes	
5	Background	Background	Describe the rationale for the review in the context of what is already	A theory of change and/or conceptual model should be presented that	Checklist	Yes	
6	Stakeholder engagement	Stakeholder engagement	The actual role of stakeholders throughout the review process (e.g. in		Checklist	No	No Stakeholder engagement
7	Objective of the review	Objective	Describe the primary question and secondary questions (when	The primary question is the main question of the review. The secondary	Checklist	Yes	
8	Definition of the question	Definition of the question	Provide reference to the question key elements, e.g. population(s).	For other question types see [4,5]	Meta-data	Population: Subtropical and Temperate Coral Systems and Coral Reefs Exposure: Ocean Warming	
9	Methods	Protocol	Provide citation, DOI or open-access link to published protocol.	The protocol should be peer-reviewed and publicly available online (open	Meta-data	10.57808/proceed.2023.9	
10		Deviations from protocol	Describe any ways in which the final methods of the review deviate from		Checklist	Yes	
11	Searches	Search strategy	Detail the search strategy used, including database names accessed.		Checklist	Yes	
12		Search string	Provide Boolean-style full search string and state the platform for which		Meta-data	Scopus: TITLE-ABS-KEY ((coral* OR "coral reef*")) AND TITLE-ABS-KEY (("ocean warming" OR "marine heatwave*" OR "marine	
13		Languages - bibliographic	List languages used in bibliographic database searches		Meta-data	English	
14		Languages - grey literature	List languages used in organisational website searches and web-based		Meta-data	English	
15		Bibliographic databases	Provide the number of bibliographic databases searched		Meta-data		2
16		Web-based search engines	Provide the number of web-based search engines searched		Meta-data		0
17		Organisational websites	Provide the number of organisational websites searched		Meta-data		1
18		Estimating comprehensiveness	Describe the process by which the comprehensiveness of the search		Checklist	Yes	
19		Search update	Describe any update to searches undertaken during the conduct of the	Compulsory (if update performed). A search update is good practice if	Checklist	Yes	
20	Article screening and study	Screening strategy	Describe the methodology for screening articles/studies for relevance.		Checklist	Yes	
21		Inclusion criteria	Describe the inclusion criteria used to assess relevance of identified		Checklist	Yes	
22	Critical appraisal	Critical appraisal strategy	Describe here the method used for critical appraisal of study validity	Optional	Checklist	No	No critical appraisal
23		Critical appraisal used in	Describe how the information from critical appraisal was used in	Compulsory if critical appraisal performed	Checklist	No	No critical appraisal
24	Meta-data extraction and	Meta-data extraction and coding	Describe the method for meta-data extraction and coding for studies.		Checklist	Yes	
25		Approaches to missing data	Describe any process for obtaining and confirming missing or unclear		Checklist	No	No missing full text
26	Data synthesis and presentation	Narrative synthesis strategy	Describe methods used for narratively synthesising the evidence base in		Checklist	Yes	
27		Knowledge gap and cluster	Describe the methods used to identify and/or prioritise key knowledge		Checklist	Yes	
28		Demonstrating procedural	Describe the role of systematic reviewers (who have also authored	Reviewers who have authored articles to be considered within the review	Checklist	Yes	
29	Results (review findings)	Description of review process	Describe the review process including the volume of evidence identified		Checklist	Yes	
30		Number of search results	Provide the number of search results from bibliographic databases	This number should not include web-based search engine or	Meta-data		1227
31		Number of search results after	Provide the total number of search results from bibliographic database	This number should not include web-based search engine or	Meta-data		91
32		Full text screening excludes	Additional file containing list of and reasons for full text exclusions.		Checklist	Yes	
33		Title screening results	Provide the number of articles retained following title screening.	Optional if screening titles and abstracts together	Meta-data	NA	
34		Abstract screening results	Provide the number of articles retained following abstract screening.	Optional if screening titles and abstracts together	Meta-data	NA	
35		Title and abstract screening	Provide the number of articles retained following title and abstract	Optional if screening titles and abstracts separately	Meta-data		187
36		Retrieval results	Provide the number of articles retrieved at full text.		Meta-data		186
37		Unobtainable articles	Additional file containing list of unobtainable articles.		Checklist	Yes	
38		Full text screening results	Provide the number of articles retained following full text screening.		Meta-data		91
39		Consistency checking: screening	Results of consistency checking at all stages (screening, meta-data		Checklist	Yes	
40		Narrative synthesis	Describe the body of evidence identified using figures and tables.		Checklist	Yes	
41		Systematic map database	Additional file containing meta-data and coding for included studies.		Checklist	Yes	
42		Limitations of the review	Discuss possible limitations in the methods used.		Checklist	Yes	
43		Limitations of the evidence base	Discuss possible limitations in the evidence base.		Checklist	Yes	
44	Conclusions	Knowledge gaps and clusters	Describe knowledge gaps (unrepresented or underrepresented subtopics		Checklist	Yes	
45		Implications for	Summarise the state of the evidence base and discuss the way in which	Reviews must not include practical environmental management	Checklist	Yes	
46		Implications for research	Discuss the way in which the identified evidence may inform research	In this section some advocacy for future research on the reviewed topic is	Checklist	Yes	
47	Declarations	Competing interests	Describe of any financial or non-financial competing interests that the		Checklist	Yes	

References

[1] James, K.L., Randall, N.P. and Haddaway, N.R., 2016. A methodology for systematic mapping in environmental sciences. *Environmental Evidence*, 5(1), p.7.

[2] Bayliss, H.R., Haddaway, N.R., Eales, J., Frampton, G.K. and James, K.L., 2016. Updating and amending systematic reviews and systematic maps in environmental management. *Environmental Evidence*, 5(1), p.20.

[3] Haddaway, N.R., Kohl, C., da Silva, N.R., Schiemann, J., Spöck, A., Stewart, R., Sweet, L.B. and Wilhelm, R., 2017. A framework for stakeholder engagement during systematic reviews and maps in environmental management. *Environmental Evidence*, 6(1), p.11.

[4] Collaboration for Environmental Evidence. 2018. Guidelines and Standards for Evidence synthesis in Environmental Management. Version 5.0. www.environmentalevidence.org/information-for-authors.

[5] Leeds Institute of Health Sciences. https://medhealth.leeds.ac.uk/info/639/information_specialists/1500/search_concept_tools. Accessed 12/11/2017.